Pranav Gangwar

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Summary

PhD Candidate seeking a full-time Spring 2027 role in computer architecture and performance modeling, with experience in GPU acceleration, kernel optimization, and parallel algorithm design across CPU, GPU, and PIM.

Education

March 2027 PhD in Computer Engineering at **University of California San Diego** (GPA: 4.0)
June 2024 MS in Computer Engineering at **University of California San Diego** (GPA: 4.0)
June 2018 BTech in Electronics and Communications Engineering at **Delhi Technological University**

Relevant Coursework: Principles of Computer Architecture, Principles of Operating Systems, Low-power VLSI for Machine Learning, Parallel Computation

Work Experience

GPU Performance Architecture Intern **Intel** (Jun-Sept 2023, Jun-Sept 2022)

- Evaluated next-generation Tensor Core architecture proposals across diverse ML workloads using an analytical performance simulator, identifying memory bottlenecks and recommending architectural improvements.
- Developed a matrix tile-size selection algorithm for Intel GPUs, improving Tensor Core utilization by reducing memory stalls and redundant computation across ML workloads.
- Optimized a state-of-the-art matrix decomposition algorithm for Intel GPUs, improving single-GPU utilization on ML workloads.
- Developed a performance model for NeRF to evaluate the proposed GPU architecture.

Digital Design Engineer **Texas Instruments** (July 2018 - July 2021)

- Led complete physical design from floorplanning to sign-off, providing architectural feedback to optimize pipelining and meet stringent I/O timing constraints.
- Reduced the leakage power of designs by resizing standard cells at the STA sign-off stage, leveraging reduced timing pessimism relative to place-and-route.

Course Projects

Fused Multi-Head Latent Attention (MLA) CUDA Kernel

- Implemented a CUDA kernel for Multi-Head Latent Attention (DeepSeek-V3-style low-rank KV-compression attention), achieving ∼**0.83x of PyTorch (cuBLAS) throughput** across **Prefill and Decode scenarios**.

GPU Accelerated General Matrix Multiplication (GEMM)

- Implemented high-performance CUDA GEMM kernels (shared-memory tiling, coalesced global memory access, register-blocked ILP), achieving ∼**0.68x of cuBLAS throughput** across **matrix sizes N=256–2049**.

Research Projects

***Computer Architecture Research* - Advisor: Tajana Rosing**

TermiNETor: An Energy-Efficient DNN Accelerator Sep 2021 - Jun 2022

- Achieved **120× energy efficiency over GPUs** by co-designing a novel bit-serial hardware architecture with an early-termination algorithm to prune redundant computations.
- Proposed the architecture and validated its performance gains using an analytical simulator, with RTL-based power and area estimation; presented at ***IEEE ICCD 2022***.

FHEmem: Near-Memory Data Processing Accelerator for FHE Sep 2021 - Dec 2022

- Co-designed a DRAM-based near-data processing (NDP) architecture that performs memory-bound FHE primitives in memory, avoiding costly off-chip data movement.

- Contributed to the near-mat unit design for in-memory number-theoretic transform (NTT) butterfly operations, enabling FHEmem to achieve **4.0x speedup and 6.9x energy-delay-area efficiency over existing accelerators**; published in ***IEEE TETC (2025 Best Paper Runner-Up)***.

Bioinformatics Research - Advisor: Yatish Turakhia

High-Resolution Variant Detection from Wastewater & Clinical Samples Jan 2023 - Jan 2026

- Developed a **scalable**, **multi-threaded**, and **memory-efficient** algorithm (WEPP) that processes millions of genome fragments against a 16-million SARS-CoV-2 genome database in hours, accurately identifying complete genomes—solving a problem previously considered intractable.
- Enabled early detection of emerging variants for timely public health response; adopted by the CDC for nationwide surveillance and published in ***PLOS Computational Biology***.
- Generalized the approach (metaWEPP) to identify complete genomes of multiple co-occurring pathogens from wastewater or clinical DNA samples; published in ***NAR Genomics and Bioinformatics***.

SARS-CoV-2 Vaccine Selector Sep 2025 - Present

- Analyzed 16 million SARS-CoV-2 genomes and found that variant prevalence and existing *in silico* fitness estimators provide complementary signals for predicting future variant dominance.
- Designed a **lightweight ML algorithm** integrating variant prevalence and fitness estimates to select vaccine variants that provide broad protection against future infections, slightly outperforming FDA recommendations.

Publications

1. **TermiNETor: Early convolution termination for efficient deep neural networks**, U Mallappa, **P Gangwar**, B Khaleghi, H Yang, T Rosing. *IEEE Int. Conference on Computer Design (ICCD)*, 2022.
2. **FHEmem: A processing in-memory accelerator for fully homomorphic encryption**, M Zhou, Y Nam, **P Gangwar**, W Xu, A Dutta, et al. *IEEE Transactions on Emerging Topics in Computing*, 2025.
3. **WEPP: Phylogenetic Placement Achieves Near-Haplotype Resolution in Wastewater-Based Epidemiology**, **P Gangwar**, P Katte, M Bhat, Y Turakhia. *PLOS Computational Biology*, 2026.
4. **metaWEPP: Leveraging Biobank-Scale Intra-Species Phylogenies for Near-Haplotype Resolution in Metagenomic Analysis**, **P Gangwar**, Q Xu, J Seangmany, P Katte, Y Turakhia. *NAR Genomics and Bioinformatics*, 2026.
5. **Genetic Diversity Drives the Rate and Fitness Jumps of Detectable SARS-CoV-2 Recombination**, K Smith, **P Gangwar**, J Wertheim, Y Turakhia. Under Review *Nature Microbiology*.

Awards & Talks

- **Best Paper Runner-Up** — FHEmem, IEEE Transactions on Emerging Topics in Computing, 2025
- **Innovation in Research Award** — Microbes in Wastewater Symposium, 2026
- **Talks** — WEPP & metaWEPP, ISMB 2026

Skills

Languages	C++, C, CUDA, Verilog, Python, Tcl, Shell
Software & Tools	PyTorch, Innovus™, Tempus™, Voltus™, PVS, Assura®, Vivado®, MATLAB®